

Appendix

Additional Related Work

Mathematical optimization methods. EVCS planning has been widely reduced to optimization problems. Li and Jenn [9] develop an integrated spatio-temporal optimization framework for charging-infrastructure planning, while Schettini et al. [18] formulate multi-year location and sizing models with multiple charging technologies and develop a specialized heuristic based on maximum-flow subproblems. He et al. [4] study public charging infrastructure planning through a contextualized location allocation model that explicitly incorporates spatial demand and planning constraints. These approaches provide a rigorous basis for infrastructure planning, but their computational requirements may become substantial as location and capacity decisions are combined with time and multiple uncertainty scenarios.

Learning-based methods. Learning-based approaches have recently been investigated to capture dynamic traffic and charging-demand patterns. Nguyen et al. [13] combine deep reinforcement learning with agent-based simulation to adapt charging station placement decisions to feedback from a dynamic urban environment. Related work uses multi-agent reinforcement learning for dynamic charging coordination and station assignment under time varying traffic conditions [16]. While these methods provide greater adaptability or modeling flexibility, they may require extensive simulation, training, or problem specific reward and model design.

Submodular optimization methods. Another line of work exploits diminishing returns in EVCS planning to obtain scalable algorithms with provable performance guarantees. Sun et al. [19] formulate early-stage charging station placement using a submodular coverage objective and develop efficient greedy optimization methods. More recently, Du et al. [3] introduce a robust link-queue-based EVCS planning model that accounts for traffic-congestion propagation and uncertain charging demand, and propose a distorted greedy algorithm by exploiting the proven submodularity of the resulting objective. Such guarantees, however, rely on preserving submodular structure and therefore do not directly extend to objectives containing complementary interactions whose marginal benefits increase with the selected set. At a more general level, objectives combining submodular and supermodular components have been studied under the BP framework [1], providing a useful perspective for modeling both diminishing return and complementary effects.

Robust and Risk-aware planning. Uncertainty in charging demand, traffic conditions, and infrastructure availability has motivated robust and scenario-based EVCS planning. Du et al. [3], for example, explicitly account for both steady and surging charging demand within a robust congestion-aware planning model. Our setting differs in that uncertainty is represented through joint demand and power grid scenarios, and adverse outcomes are explicitly controlled through a Conditional Value-at-Risk (CVaR) term [17]. This allows the planning objective to consider not only expected performance but also the quality of solu-

tions in the unfavorable tail of the scenario distribution. Combining CVaR with discrete diminishing returns objectives is known to be hard: Maehara [11] shows that the CVaR of a stochastic submodular function is not submodular and admits no multiplicative approximation by a single set. Approximation results can be recovered only under relaxed formulations or continuous-domain variants [21]. We keep the single set formulation required by a deployable investment plan and accept that a worst case guarantee is unavailable in exchange.

Existing EVCS planning studies often treat location selection, sizing, dynamic demand, congestion, and uncertainty separately. We address this limitation by developing a unified time-aware and risk-aware budgeted planning model that jointly captures discrete capacities, heterogeneous costs, grid stress, corridor synergy, and tail risk, and optimize it using *Scenario-Based Global Trajectory Optimization* (SGTO).

Problem Formulation

We use three standard closure properties [7]: (P1) a nonnegative combination of submodular functions is submodular, and $-F$ is supermodular if F is submodular; (P2) for nonnegative modular g and nondecreasing φ , $\varphi \circ g$ is submodular if φ is concave and supermodular if convex; (P3) the weighted coverage $X \mapsto \sum_u d_u [1 - \prod_{e \in X} (1 - a_{u,e})]$ with $a_{u,e} \in [0, 1]$ is monotone submodular, and $X \mapsto \sum_{e < f \in X} b_{ef}$ with $b_{ef} \geq 0$ is monotone supermodular. For a random loss L , the *conditional value-at-risk* at level $\delta \in (0, 1)$ is the expected loss in the worst $(1 - \delta)$ tail, with variational form [17]

$$\text{CVaR}_\delta(L) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{1-\delta} \mathbb{E}[(L - \tau)^+] \right\}, \quad (10)$$

computable by sorting scenario losses for a discrete scenario distribution.

Structural Properties of the Global Reward

The global reward function. Let $\mathcal{T} = \{1, \dots, T\}$ be the planning periods and $\Omega = \{\omega_1, \dots, \omega_M\}$ a set of demand and power grid scenarios with probabilities p_ω (e.g. weekday, weekend, peak, event surge, reduced grid capacity). Under scenario ω , $d_{u,t}^\omega$ is the charging demand of region $u \in \mathcal{U}$ in period t and $g_{z,t}^\omega$ the available capacity of grid region $z \in \mathcal{Z}$. The **global reward** of $X \in \mathcal{F}$ is

$$F_\omega(X) = \alpha C_\omega(X) + \beta Y(X) - \gamma P_\omega(X) - \eta U_\omega(X), \quad (11)$$

with time-dependent coverage C_ω , route synergy Y , grid overload penalty P_ω , unmet-demand penalty U_ω and weights $\alpha, \beta, \gamma, \eta \geq 0$; the expected reward is $\bar{F}(X) = \sum_\omega p_\omega F_\omega(X)$.

Coverage. With a distance-decay kernel $\phi(u, i) \in [0, 1]$, mean hourly demand $\bar{d}_u$ and attracted demand $\text{att}(i) = \sum_u \phi(u, i) \bar{d}_u$, the effectiveness of $e = (i, \ell)$ is $a_{u,e} = \phi(u, i) \sigma_\ell \min(1, \bar{\zeta} q_e / \text{att}(i))$, with level factor $\sigma_\ell \in (0, 1]$ and reference utilization $\bar{\zeta}$: a small station in a dense area is congested, which makes the capacity decision non-trivial. Then $C_\omega(X) = \sum_t w_t \sum_u d_{u,t}^\omega [1 - \prod_{e \in X} (1 - a_{u,e})]$

with period weights $w_t \geq 0$ counts overlapping coverage once and is monotone submodular by (P3), (P1).

Synergy. Two configurations form a corridor, $b_{ef} = \kappa \geq 0$, if their locations are adjacent on the road network and within road distance $D_{\max}$, and $b_{ef} = 0$ otherwise; $Y(X) = \sum_{e < f \in X} b_{ef}$ is monotone supermodular by (P3), and stations on the same route but far apart earn no artificial synergy.

Grid overload. With background load $\ell_{z,t}^\omega$, utilization $\zeta_{e,t}^\omega \in [0, 1]$ and $\mathcal{E}_z$ the configurations connected to region z ,

$$P_\omega(X) = \sum_{t \in \mathcal{T}} \sum_{z \in \mathcal{Z}} \left(\left[\ell_{z,t}^\omega + \sum_{e \in X \cap \mathcal{E}_z} \zeta_{e,t}^\omega q_e - g_{z,t}^\omega \right]^+ \right)^2, \quad (12)$$

a convex nondecreasing function of a modular load, hence supermodular by (P2), (P1).

Unmet demand. Served demand is $s_{u,t}^\omega(X) = \min\{d_{u,t}^\omega, \nu \zeta_t^\omega \sum_{e \in X} a_{u,e} q_e\}$ with efficiency $\nu > 0$, assuming effective capacity aggregates additively before the demand cap. It is concave in a modular function, hence submodular by (P2), so $U_\omega(X) = \sum_{t,u} (d_{u,t}^\omega - s_{u,t}^\omega(X))$ is supermodular by (P1).

Structure. Coverage has diminishing returns while corridor synergy has increasing returns, so no single curvature class represents all four terms. Grouping terms by (P1)–(P2),

$$\bar{F}(X) = \underbrace{\sum_\omega p_\omega (\alpha C_\omega - \gamma P_\omega - \eta U_\omega)(X)}_{\text{submodular}} + \underbrace{\beta Y(X)}_{\text{supermodular}}. \quad (13)$$

To penalize poor performance in adverse scenarios, we solve $\max_{X \in \mathcal{F}} F^{\text{rob}}(X)$, where

$$F^{\text{rob}}(X) = \bar{F}(X) - \rho \text{CVaR}_\delta(L_\omega(X)), \quad (14)$$

where $\rho \geq 0$ is the risk aversion and $L_\omega := \gamma P_\omega + \eta U_\omega$ the scenario loss. Eq. (13) does *not* extend to F^{rob} : for fixed τ , $X \mapsto \mathbb{E}[(L_\omega(X) - \tau)^+]$ is supermodular by (P2), (P1), but CVaR is the pointwise minimum over τ in Eq. (10), which need not be supermodular; in general no single set approximates the optimal CVaR within any multiplicative factor [11], and guarantees are recovered only for portfolios or continuous relaxations [15,21]. Moreover, the submodular group in Eq. (13) is not monotone, whereas [1] assumes monotone components. SGTO therefore borrows the structure of BP methods, greedy construction followed by semi-gradient linearization [1,12,5], but inherits no approximation guarantee; improvement is enforced by a validation gate.

Additional Experiment

Table 3. Budget sweep (12 test scenarios). SA and SGTO: mean $\pm$ std over 3 seeds; greedy baselines are deterministic. All methods spend the full budget.

Method	$B = 20,000$	$B = 30,000$	$B = 40,000$
Simulated Annealing	1780.8 ± 68.5	2121.9 ± 23.6	2312.0 ± 41.1
Cost-Aware Greedy	1952.9	2277.9	2281.5
Greedy + One-Exchange	1960.1	2338.4	2326.7
SGTO	1962.5 ± 4.2	2338.3 ± 14.4	2336.3 ± 29.1

Table 4. Results on the enlarged held-out set of 32 test days (training and validation unchanged); mean $\pm$ std over 10 seeds for SA and SGTO.

Method	ΔF^{rob}	F^{worst}	CVaR loss	FR
Simulated Annealing	2021.3 ± 35.8	-2140.1 ± 48.0	2677.7 ± 38.0	0.483 ± 0.006
Cost-Aware Greedy	2051.5	-2117.8	2721.1	0.481
Greedy + One-Exchange	2087.8	-2040.2	2667.1	0.477
SGTO	2165.2 ± 51.0	-1861.9 ± 109.3	2582.4 ± 62.5	0.468 ± 0.007